Bayesian Optimization Report: Function 2

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.724	Good
2	0.724	Good
3	0.723	Good
4	0.740	Good
5	0.740	Good
6	0.724	Good
7	0.723	Good
8	0.736	Good
9	0.721	Good
10	0.721	Good

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.1533	0.0569	0.2245
2	0.1386	0.0561	0.1942
3	0.1305	0.0551	0.1877
4	0.1212	0.0771	0.1697
5	0.1074	0.0865	0.1328
6	0.1159	0.0633	0.1578
7	0.1120	0.0638	0.1532
8	0.1004	0.0746	0.1302
9	0.1083	0.0578	0.1469
10	0.1047	0.0587	0.1445

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.23877715 0.11047411]	0.053520333810452014	Matern(ν=0.5)	UCB	0.724	None

2	[0.78057593 0.91615942]	0.33843875205245105	Matern(ν=0.5)	UCB	0.724	None
3	[0.07568966 0.18390212]	0.021099785579781094	Matern(ν=0.5)	UCB	0.723	None
4	[0.50641073 0.60614201]	0.14908912847002773	Matern(ν=1.5)	UCB	0.740	None
5	[0.689732 0.01072249]	0.6196843294338252	Matern(ν=1.5)	UCB	0.740	None
6	[0.75259565 0.01751269]	0.40035737263119386	Matern(ν=0.5)	UCB	0.724	None
7	[0.63120209 0.37477238]	0.31879783811134127	Matern(ν=0.5)	UCB	0.723	None
8	[0.69523499 0.14196323]	0.637374218280027	Matern(ν=0.5)	UCB	0.736	None
9	[0.68136379 0.96412922]	0.5996662787219182	Matern(ν=0.5)	UCB	0.721	None
10	[0.68607217 0.63759914]	0.6247145339536837	Matern(ν=0.5)	UCB	0.721	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	0.699679
EI	0.008924
PI	0.199715
THOMPSON	0.622040
KG	0.008924
ENTROPY	1.139166
PORTFOLIO	0.770333

Iteration 2

Method	Max Value
UCB	0.566224
EI	0.008146
PI	0.173126
THOMPSON	0.575923
KG	0.008146
ENTROPY	1.091689

PORTFOLIO	0.686729

Iteration 3

Method	Max Value
UCB	0.565373
EI	0.007799
PI	0.188395
THOMPSON	0.564213
KG	0.007799
ENTROPY	1.031713
PORTFOLIO	0.586269

Iteration 4

Method	Max Value
UCB	0.558183
EI	0.004580
PI	0.105682
THOMPSON	0.537183
KG	0.004580
ENTROPY	1.018986
PORTFOLIO	0.548825

Iteration 5

Method	Max Value
UCB	0.537888
EI	0.002402
PI	0.060772
THOMPSON	0.658005
KG	0.002402
ENTROPY	1.022737
PORTFOLIO	0.526662

Iteration 6

Method	Max Value
UCB	0.540705

EI	0.003312
PI	0.089782
THOMPSON	0.508179
KG	0.003312
ENTROPY	0.971689
PORTFOLIO	0.499007

Iteration 7

Method	Max Value
UCB	0.540550
EI	0.003164
PI	0.091903
THOMPSON	0.544907
KG	0.003164
ENTROPY	0.993951
PORTFOLIO	0.461221

Iteration 8

Method	Max Value
UCB	0.537559
EI	0.002225
PI	0.062694
THOMPSON	0.572906
KG	0.002225
ENTROPY	1.030913
PORTFOLIO	0.466487

Iteration 9

Method	Max Value
UCB	0.529851
EI	0.001848
PI	0.064176
THOMPSON	0.593922
KG	0.001848
ENTROPY	1.015000

PORTFOLIO	0.452526

Iteration 10

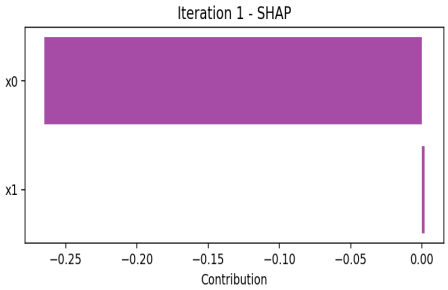
Method	Max Value
UCB	0.524242
EI	0.001392
PI	0.051376
THOMPSON	0.555637
KG	0.001392
ENTROPY	0.937021
PORTFOLIO	0.430899

LOO MSE Summary

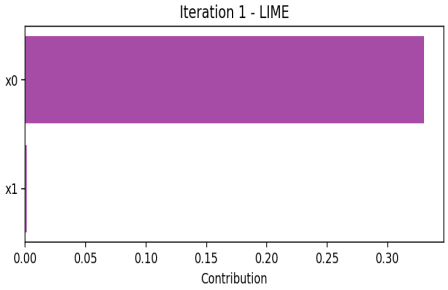
Iteration	LOO MSE
1	0.022323
2	0.020580
3	0.018926
4	0.022018
5	0.016678
6	0.014698
7	0.013940
8	0.012696
9	0.012672
10	0.011543

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



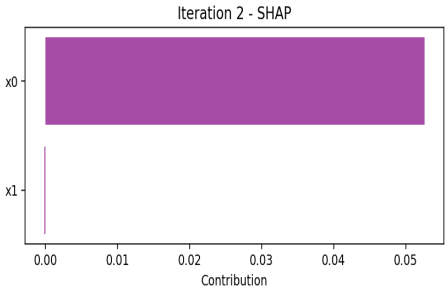
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	-0.26491
x1	0.00189

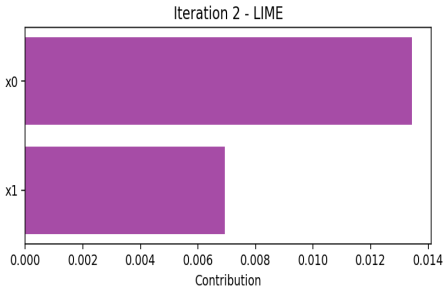
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.33010
x1	0.00119

Iteration 2 - SHAP



Iteration 2 - LIME



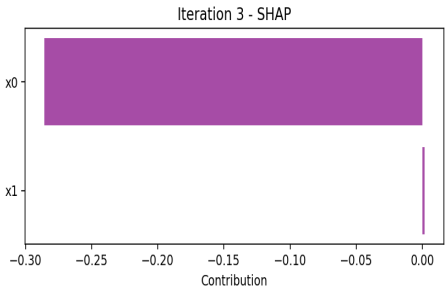
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	0.05264
x1	-0.00014

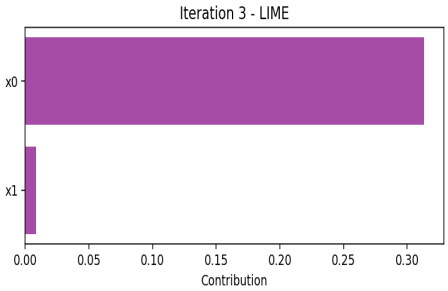
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.01344
x1	0.00692

Iteration 3 - SHAP



Iteration 3 - LIME



Iteration 3 - SHAP Values

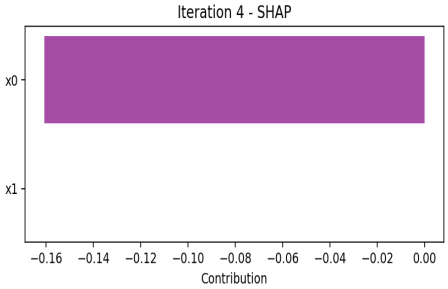
Feature	SHAP Value
x0	-0.28569
x1	0.00183

Iteration 3 - LIME Values

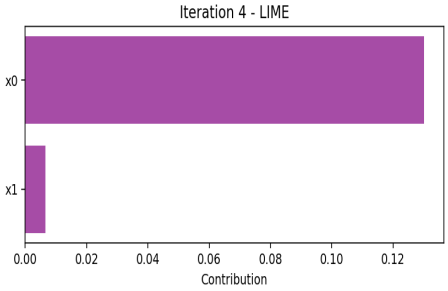
Feature	LIME Value
x0	0.31307

x1	0.00843

Iteration 4 - SHAP



Iteration 4 - LIME



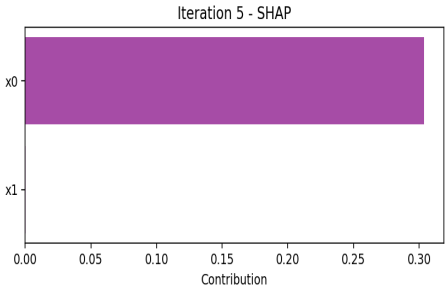
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	-0.16080
x1	0.00002

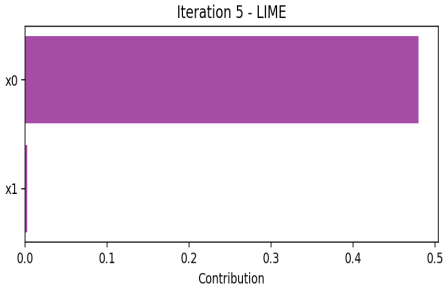
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.13006
x1	0.00655

Iteration 5 - SHAP



Iteration 5 - LIME



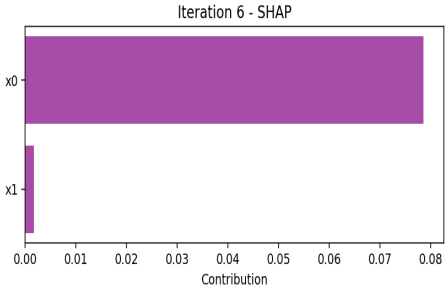
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	0.30366
x1	0.00018

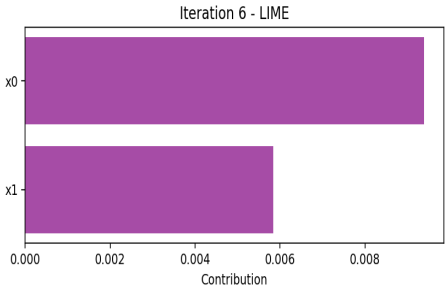
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.48003
x1	0.00209

Iteration 6 - SHAP



Iteration 6 - LIME



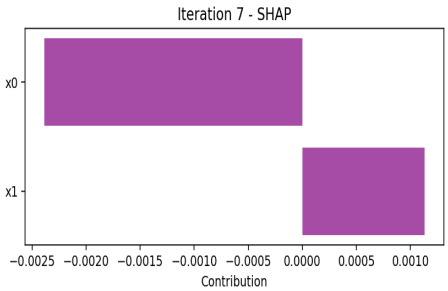
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	0.07863
x1	0.00173

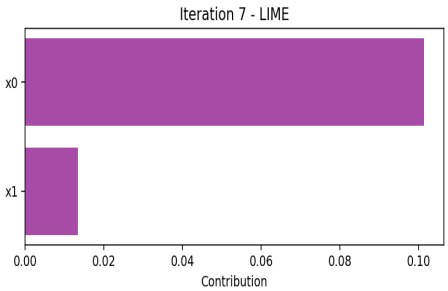
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.00938
x1	0.00584

Iteration 7 - SHAP



Iteration 7 - LIME



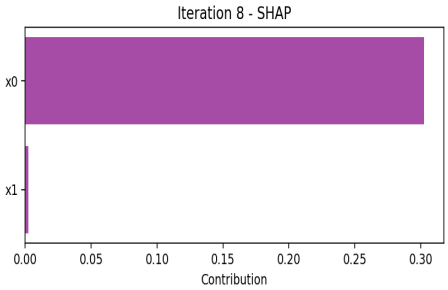
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	-0.00239
x1	0.00113

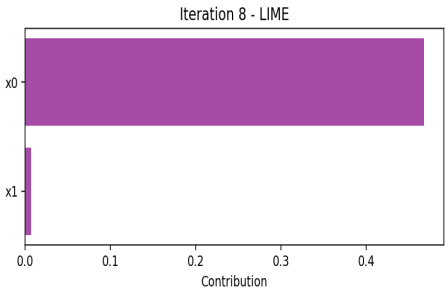
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.10129
x1	0.01337

Iteration 8 - SHAP



Iteration 8 - LIME



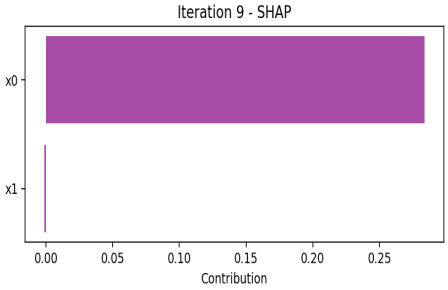
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.30241
x1	0.00249

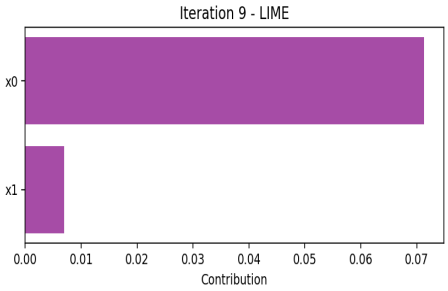
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.46748
x1	0.00702

Iteration 9 - SHAP



Iteration 9 - LIME



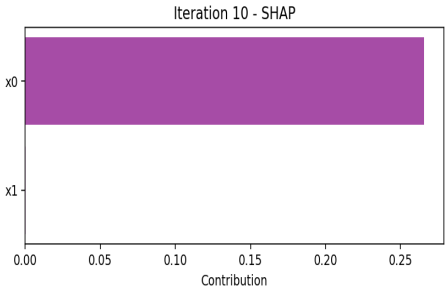
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	0.28397
x1	-0.00120

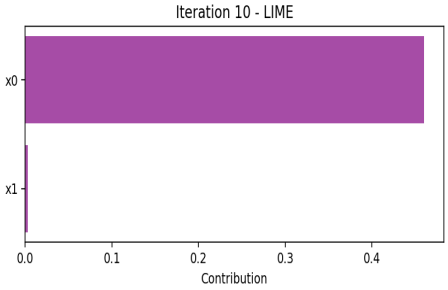
Iteration 9 - LIME Values

Feature	LIME Value
x0	0.07129
x1	0.00693

Iteration 10 - SHAP



Iteration 10 - LIME



Iteration 10 - SHAP Values

Feature	SHAP Value
x0	0.26551
x1	0.00024

Iteration 10 - LIME Values

Feature	LIME Value
x0	0.46001
x1	0.00311